

X(6th Sm.)-Computer Sc.-H/CC-14/CBCS

2022

COMPUTER SCIENCE — HONOURS

Paper : CC-14

(Theory of Computation)

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions :

2×5

- (a) When is a string y said to be accepted by a finite automaton M ?
- (b) Prove that if δ is a transition function such that $\delta(q, w) = \delta(q, x)$, then $\delta(q, wy) = \delta(q, xy)$.
- (c) Draw a transition diagram for a DFA M that accepts the string 101101 over $\Sigma = \{0, 1\}$.
- (d) Define Push-down Automata.
- (e) Give a context-free grammar of palindrome.
- (f) What is Context Sensitive Grammar?
- (g) Draw the transition diagram of a FA which accepts the strings having any number of a 's (possibly ϵ) followed by any number of b 's (possibly ϵ) in $\Sigma = \{a, b\}$. Write the regular expression also.
- (h) State one similarity and one dissimilarity between a Turing machine and a General purpose computer.

2. (a) Convert the following Mealy machine M to an equivalent Moore machine. Show the steps clearly.

Present State	Next State			
	$a = 0$		$a = 1$	
	State	Output	State	Output
$\rightarrow q_1$	q_1	1	q_2	0
q_2	q_4	1	q_4	1
q_3	q_2	1	q_3	1
q_4	q_3	0	q_1	1

(b) Find the language generated by the following grammar : $S \rightarrow 0A|1S|0|1$, $A \rightarrow 1A|1S|1$ 5+5

Please Turn Over

3. (a) Why is a NDFA called so? Draw the transition diagram of a NDFA that accepts all strings whose second last symbol is b where $\Sigma = \{a, b\}$.

(b) Construct a DFA from the NDFA constructed in the previous part [i.e. 3(a)]. (1+4)+5

4. (a) Construct a grammar that accepts the following set.

$$\{0^n 1^m 0^n \mid m, n \geq 1\}$$

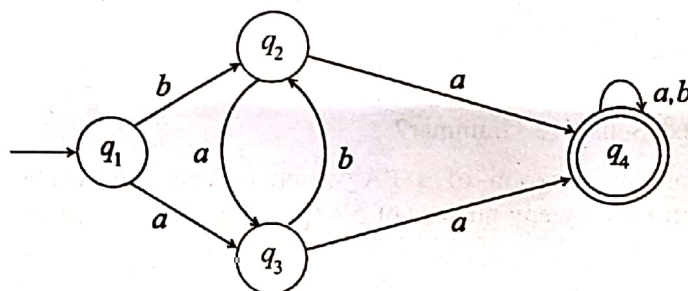
- (b) Write down the steps to convert a non-deterministic finite automata into equivalent deterministic finite automata. 5+5

5. (a) Find the language generated by the grammar $S \rightarrow AB, A \rightarrow A1|0, B \rightarrow 2B|3$.
Can the above language be generated by a grammar of higher type?

- (b) Design a Turing machine over $\{1, b\}$ which can compute concatenation function over $\Sigma = \{1\}$.
If a pair of words (w_1, w_2) is the input, the output has to be $w_1 w_2$. 5+5

6. (a) Write the regular expressions (over $\Sigma = \{a, b\}$), for the case when no two a 's or no two b 's appear together.

- (b) Use Arden's method to find the regular expression from the following DFA. Show the steps. 5+5



7. (a) Prove the following identity :

$$(a^*ab + ba)^*a^* = (a + ab + ba)^*$$

- (b) In a context free grammar what is a null production? Give an example.

- (c) When is a variable A in a context free grammar said to be nullable? 6+(2+1)+1

8. (a) Let $G = (\{A, B, S\}, \{0, 1\}, P, S)$, where P consists of $S \rightarrow 0AB, A0 \rightarrow S0B, A1 \rightarrow SB1, B \rightarrow SA, B \rightarrow 01$. Find the language generated.

- (b) State the significance of the Halting problem in Turing machine.

- (c) Design a Turing machine to recognize all strings containing even number of 1's. 5+2+3